

● MEDIUM

● CRAFT AND STRUCTURE

Cross-Text Connections

10 questions · ~15 min

02

Text 1

Little is known about how plate tectonics—wherein slabs of Earth’s crust move over, under, away from, and against one another—began. Some researchers contend that tectonic movements began around 3 billion years ago, often noting that computer models of Earth’s mantle temperature at the time indicate that the mantle would have been sufficiently molten to enable the plates to move.

Text 2

Ultimately, any plausible claim about the inception of tectonic movement must rest on empirical evidence from the geological record. Researcher Wriju Chowdhury and his team analyzed the geochemistry of zircon crystals to gain insight into the chemical composition of the magma from which the crystals formed and, based on the data, compellingly argue that plate tectonics may have been occurring as early as 4.2 billion years ago.

Based on the texts, how would the author of Text 2 most likely respond to what “some researchers contend” as described in Text 1?

- A. By suggesting that the temperature of Earth’s mantle 3 billion years ago was likely insufficient to allow for the level of tectonic movement predicted by computer models
- B. By distinguishing between computer models of Earth’s mantle temperature that reliably predict the onset of plate tectonics and those that do not
- C. By indicating that computer models of Earth’s mantle temperature are still being improved such that new models tend to be much more reliable than their predecessors

- D. By asserting that a more definitive form of evidence than the computer models suggests a different timeline for the onset of plate tectonics on Earth

Text 1

French Impressionist artist Edgar Degas insisted that his paintings be kept in their original frames after they were sold. Like many Impressionist painters, Degas used painted frames that stood in contrast to the gold frames frequently seen at the Paris Salon, a prestigious art exhibition that was sponsored by the French government and promoted traditional painting styles. Impressionist painters likely chose these colorful frames to distinguish themselves from what was considered conventional at the time.

Text 2

Impressionist painters often focused on the interplay of color and light in their works. As such, those Impressionists who placed their works in painted frames instead of the more traditional gold ones did so for aesthetic reasons: a frame's color was likely chosen because it would harmonize with the colors or subjects in a painting. Gold, conversely, could distract from the subtleties in a painted scene.

Based on the texts, both authors would most likely agree with which statement?

- A. Gold frames were considered especially desirable by those who purchased works from Impressionist painters.
- B. The colors in an Impressionist painting were often chosen to complement the colors of the frame it would be placed in.
- C. Many Impressionist painters were intentional about the frames they selected for their works.
- D. Degas's preferred framing style was different from that of most Impressionist painters.

Text 1

In a study of the benefits of having free time, Marissa Sharif found that the reported sense of life satisfaction tended to plateau when participants had two hours of free time per day and actually began to fall when they had five hours of free time per day. After further research, Sharif concluded that this dip in life satisfaction mainly occurred when individuals spent all their free time unproductively, such as by watching TV or playing games.

Text 2

Psychologist James Maddux cautions against suggesting an ideal amount of free time. The human desire for both free time and productivity is universal, but Maddux asserts that individuals have unique needs for life satisfaction. Furthermore, he points out that there is no objective definition for what constitutes productivity; reading a book might be considered a productive activity by some, but idleness by others.

Based on the texts, how would Maddux (Text 2) most likely respond to the conclusion Sharif (Text 1) reached after her further research?

- A. By acknowledging that free time is more likely to enhance life satisfaction when it is spent productively than when it is spent unproductively
- B. By challenging the reasoning in Text 1, as it has not been proved that productivity commonly contributes to individuals' life satisfaction
- C. By warning against making an overly broad assumption, as there is no clear consensus in distinguishing between productive and unproductive activities

- D. By claiming that the specific activities named in Text 1 are actually examples of productive activities rather than unproductive ones

Text 1

In 2021, a team led by Amir Siraj hypothesized that the Chicxulub impactor—the object that struck the Yucatán Peninsula sixty-six million years ago, precipitating the mass extinction of the dinosaurs—was likely a member of the class of long-period comets. As evidence, Siraj cited the carbonaceous chondritic composition of samples from the Chicxulub impact crater as well as of samples obtained from long-period comet Wild 2 in 2006.

Text 2

Although long-period comets contain carbonaceous chondrites, asteroids are similarly rich in these materials. Furthermore, some asteroids are rich in iridium, as Natalia Artemieva points out, whereas long-period comets are not. Given the prevalence of iridium at the crater and, more broadly, in geological layers deposited worldwide following the impact, Artemieva argues that an asteroid is a more plausible candidate for the Chicxulub impactor.

Based on the texts, how would Artemieva likely respond to Siraj’s hypothesis, as presented in Text 1?

- A. By insisting that it overestimates how representative Wild 2 is of long-period comets as a class
- B. By arguing that it does not account for the amount of iridium found in geological layers dating to the Chicxulub impact
- C. By praising it for connecting the composition of Chicxulub crater samples to the composition of certain asteroids
- D. By concurring that carbonaceous chondrites are prevalent in soil samples from sites distant from the Chicxulub crater

Text 1

When companies in the same industry propose merging with one another, they often claim that the merger will benefit consumers by increasing efficiency and therefore lowering prices. Economist Ying Fan investigated this notion in the context of the United States newspaper market. She modeled a hypothetical merger of Minneapolis-area newspapers and found that subscription prices would rise following a merger.

Text 2

Economists Dario Focarelli and Fabio Panetta have argued that research on the effect of mergers on prices has focused excessively on short-term effects, which tend to be adverse for consumers. Using the case of consumer banking in Italy, they show that over the long term (several years, in their study), the efficiency gains realized by merged companies do result in economic benefits for consumers.

Based on the texts, how would Focarelli and Panetta (Text 2) most likely respond to Fan's findings (Text 1)?

- A. They would recommend that Fan compare the near-term effect of a merger on subscription prices in the Minneapolis area with the effect of a merger in another newspaper market.
- B. They would argue that over the long term the expenses incurred by the merged newspaper company will also increase.
- C. They would encourage Fan to investigate whether the projected effect on subscription prices persists over an extended period.
- D. They would claim that mergers have a different effect on consumer prices in the newspaper industry than in most other industries.

Text 1

Stage lighting theorist Adolphe Appia was perhaps the first to argue that light must be considered alongside all the various elements of a stage to create a single, unified performance. Researcher Kelly Bremner, however, has noted that Appia lacked technical expertise in the use of light in the theater. As a result of Appia's inexperience, Bremner argues, Appia's theory of light called for lighting practices that weren't possible until after the advent of electricity around 1881.

Text 2

Adolphe Appia was not an amateur in the practice of lighting. Instead, it is precisely his exposure to lighting techniques at the time that contributed to his theory on the importance of light. When working as an apprentice for a lighting specialist in his youth, Appia observed the use of portable lighting devices that could be operated by hand. This experience developed his understanding of what was possible in the coordination of elements on the stage.

Based on the texts, how would the author of Text 2 most likely respond to the claim about Appia's level of technical expertise made by Bremner in Text 1?

- A. Many lighting technicians dismissed Appia's ideas about light on the stage.
- B. Appia likely gained a level of technical expertise during his time as an apprentice.
- C. Theater practitioners who worked with Appia greatly admired his work.
- D. Appia was unfamiliar with the use of music and sound in theater.

Text 1

Polar bears sustain themselves primarily by hunting seals on the Arctic sea ice, but rising ocean temperatures are causing the ice to diminish, raising concerns about polar bear population declines as these large predators' seal-hunting habitats continue to shrink. A 2020 study examining polar bear populations across the Arctic concluded that populations affected by sea-ice loss are at great risk of extinction by the end of the twenty-first century.

Text 2

Monitoring carried out by researchers from the Norwegian Polar Institute shows that the polar bear population on the Arctic archipelago of Svalbard remains stable and well nourished despite rapidly declining sea ice in recent years. The researchers attribute this population's resilience in part to a shift in feeding strategies: in addition to hunting seals, the Svalbard polar bears have begun relying on a diet of reindeer meat and birds' eggs.

Based on the texts, how would the researchers in Text 2 most likely respond to the conclusion presented in the underlined portion of Text 1?

- A. By noting that it neglects the possibility of some polar bear populations adapting to changes in their environment
- B. By suggesting that it is likely incorrect about the rates at which warming ocean temperatures have caused sea ice to melt in the Arctic
- C. By asserting that it overlooks polar bear populations that have not yet been affected by loss of seal-hunting habitats
- D. By arguing that it fails to account for polar bears' reliance on a single seal-hunting strategy

Text 1

A tiny, unusual fossil in a piece of 99-million-year-old amber is of the extinct species *Oculudentavis khaungraae*. The *O. khaungraae* fossil consists of a rounded skull with a thin snout and a large eye socket. Because these features look like they are avian, or related to birds, researchers initially thought that the fossil might be the smallest avian dinosaur ever found.

Text 2

Paleontologists were excited to discover a second small fossil that is similar to the strange *O. khaungraae* fossil but has part of the lower body along with a birdlike skull. Detailed studies of both fossils revealed several traits that are found in lizards but not in dinosaurs or birds. Therefore, paleontologists think the two creatures were probably unusual lizards, even though the skulls looked avian at first.

Based on the texts, what would the paleontologists in Text 2 most likely say about the researchers' initial thought in Text 1?

- A. It is understandable because the fossil does look like it could be related to birds, even though *O. khaungraae* is probably a lizard.
- B. It is confusing because it isn't clear what caused the researchers to think that *O. khaungraae* might be related to birds.
- C. It is flawed because the researchers mistakenly assumed that *O. khaungraae* must be a lizard.
- D. It is reasonable because the *O. khaungraae* skull is about the same size as the skull of the second fossil but is shaped differently.

Text 1

In 1954 George Balanchine choreographed a production of *The Nutcracker*, a ballet by Pyotr Ilyich Tchaikovsky. It has since become a tradition for hundreds of dance companies in North America to stage *The Nutcracker* each year. But the show is stuck in the past, with an old-fashioned story and references, so it should no longer be produced. Ballet needs to create new traditions if it wants to stay relevant to contemporary audiences.

Text 2

The Nutcracker is outdated, but it should be kept because it's a holiday favorite and provides substantial income for some dance companies. Although it can be behind the times, there are creative ways to update the show. For example, Debbie Allen successfully modernized the story. Her show *Hot Chocolate Nutcracker* combines ballet, tap, hip-hop, and other styles, and it has been gaining in popularity since it opened in 2009.

Based on the texts, how would the author of Text 2 most likely respond to the underlined claim in Text 1?

- A. By questioning the idea that the story of *The Nutcracker* is stuck in the past and by rejecting the suggestion that contemporary audiences would enjoy an updated version
- B. By agreeing that contemporary audiences have largely stopped going to see performances of *The Nutcracker* because it's so old-fashioned
- C. By pointing out that most dance companies could increase their incomes by offering modernized versions of *The Nutcracker*

- D. By suggesting that dance companies should consider offering revised versions of *The Nutcracker* instead of completely rejecting the show

Text 1

It seems clear that emotional contagion (the unintentional transfer of an emotional state from one person to another) requires physical interaction and the observation of body language. After all, research shows that talking to someone who is smiling and expressing positive feelings often causes people to respond in a comparably positive way. Similarly, displays of nervous fidgeting have been found to prompt others to begin behaving more nervously, too.

Text 2

In an experiment using a social networking service, Zeyao Yang and Emilio Ferrara found evidence of emotional contagion in text-based online interactions. The researchers discovered that reading social media posts that expressed a positive outlook led people to make more positive posts themselves, while posts with a negative emotional tone led people to make more negative posts.

Based on the texts, what would the researchers in Text 2 most likely say about the claim underlined in Text 1?

- A. It perpetuates a flawed understanding of emotional contagion, because there isn't enough evidence to suggest that smiling is a sign of emotional contagion.
- B. It reflects an incomplete view of emotional contagion, because this phenomenon can occur even without in-person interaction.
- C. It's fairly persuasive, because studies attempting to identify emotional contagion in situations without in-person interaction have thus far yielded unclear results.
- D. It's mostly accurate, because the social networking study confirmed that emotional contagion primarily occurs in response to negative emotions like nervousness.